

1.5T TORSO ARRAY COIL (T3014CA) TROUBLESHOOTING GUIDE

Section I: Checking the PIN diodes with Digital Multimeter (DMM)

- 1) Remove the replaceable covers.
- 2) Remove the covers over the circuit boards to gain access to the PIN diodes (See Diagram 1 below).

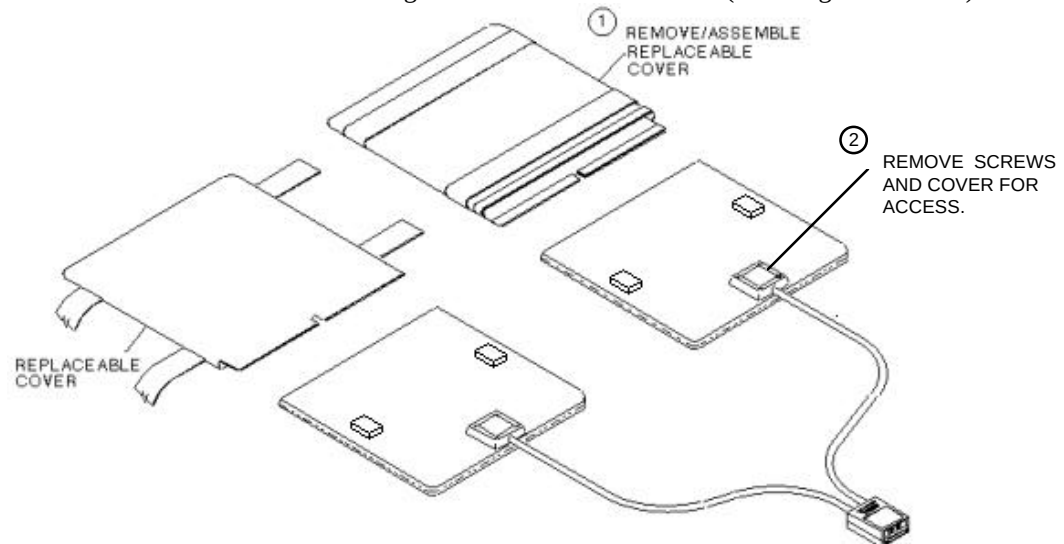


DIAGRAM 1

- 3) Select the DIODE TEST function on the Digital Multimeter (DMM)
- 4) Perform forward/reverse diode check on all 4 diodes. There are 2 diodes in the posterior coil, and 2 in the anterior coil.

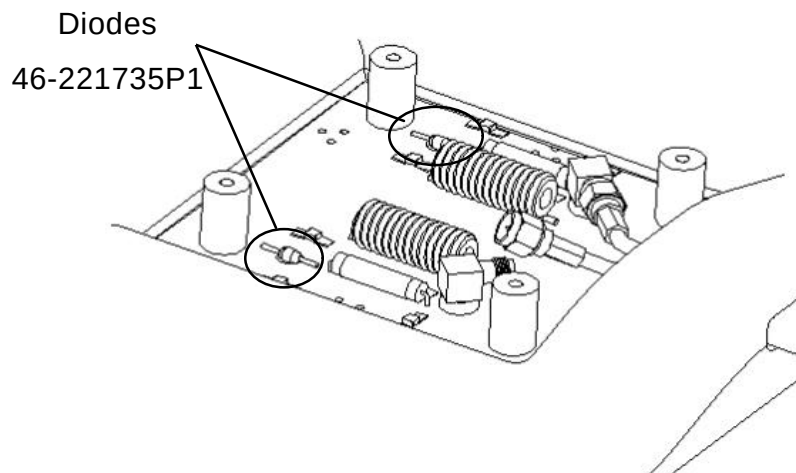


DIAGRAM 2

- 5) Compare your readings to the following specs.....
 - a.) A reading of **0.400 to 0.600** should be observed on the DMM in **Forward bias**.
 - b.) A reading of **OL** should be observed on the DMM in **Reverse bias**. Any other readings indicate a bad diode.
- 6) If an open or shorted condition exists from the readings in Step 5, **REPLACE THE DIODES WITH 9415's, PART NUMBER 46-221735P1.**

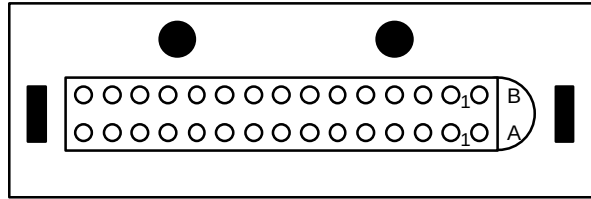
CAUTION:

USE A 25W SOLDER IRON AND PROPER ESD PROTECTION. BE CAREFUL NOT TO DAMAGE THE CIRCUIT BOARD OR THE DIODE BY OVERHEATING THE SOLDER.

- 7) If diodes were replaced, repeat Steps 4 and 5 above.

Section II: Checking the External Cable

- 1) Remove the replaceable covers.
- 2) Remove the covers over the circuit boards to gain access to the cable connections (See Diagram 1 in Section I).



**PIN LOCATIONS ON EXTERNAL CONNECTOR
DIAGRAM 3**

- 3) Using the DMM, Ohm the **Center Pin** of the cable of **ANTERIOR** coil left to **Pin B5** on the Quick Disconnect Box. Continuity should be read on the meter. Flex the appropriate cable, especially near the connectors and the strain relief and observe that continuity is maintained with no instability or fluctuations. Continue this method for steps 4-11.
- 4) Ohm **Center Pin** of **ANTERIOR** coil right to **Pin B7** on the Quick Disconnect Box.
- 5) Ohm **Center Pin** of **POSTERIOR** coil left to **Pin B9** on the Quick Disconnect Box.
- 6) Ohm **Center Pin** of **POSTERIOR** coil right to **Pin B11** on the Quick Disconnect Box.
- 7) Ohm **Ground** of **ANTERIOR** coil left to **Pin A1** on the Quick Disconnect Box.
- 8) Ohm **Ground** of **ANTERIOR** coil right to **Pin A1** on the Quick Disconnect Box.
- 9) Ohm **Ground** of **POSTERIOR** coil left to **Pin A1** on the Quick Disconnect Box.
- 10) Ohm **Ground** of **POSTERIOR** coil right to **Pin A1** on the Quick Disconnect Box.
- 11) If an open condition exists in any of the steps above, **REMOVE THE COVER FROM THE QUICK DISCONNECT BOX AND CHECK THE SMB CONNECTIONS TO THE PHASE ADJUST BD.** These connections have been known to come loose due to twisting of the cables. If one was loose, reseal the connection and check continuity again as well as any suspect diode readings you may have obtained. **IF THEY ARE TIGHT, REPLACE THE COIL (T3014CA). THE EXTERNAL CABLES ARE NOT FRU'S.**

TROUBLESHOOTING STEPS IN REVIEW:

- 1) **CHECK THE DIODES.**
- 2) **CHECK THE CABLE.**
- 3) **CHECK THE SMB CONNECTIONS INSIDE THE QUICK DISCONNECT BOX.**
- 4) **IF THE DIODES ARE BAD, REPLACE THEM WITH 9415s (46-221735P1).**
- 5) **IF THE CABLE IS BAD, REPLACE THE COIL (T3014CA).**